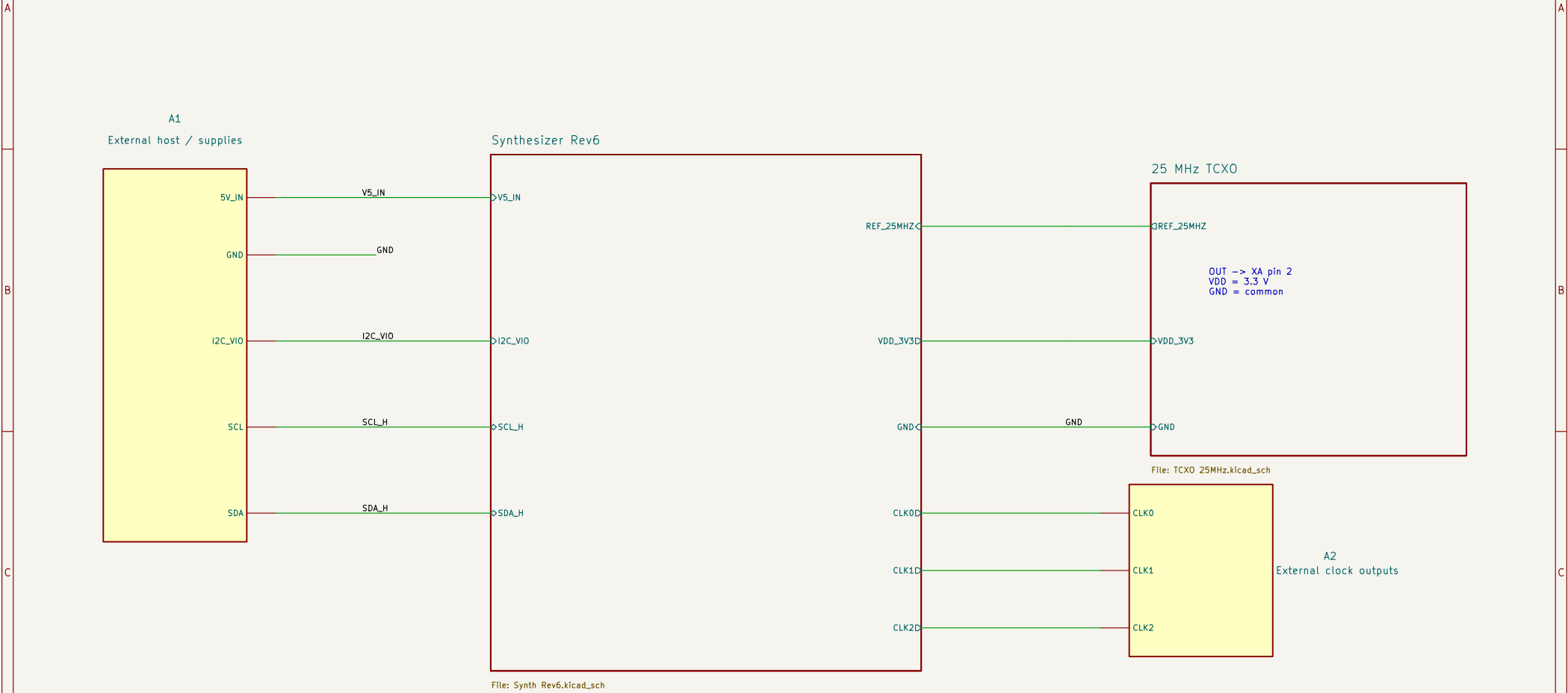


QRP LABS MODULES | 25 MHz TCXO + SYNTH Rev6

Intended assembly interconnection of two separately photographed boards



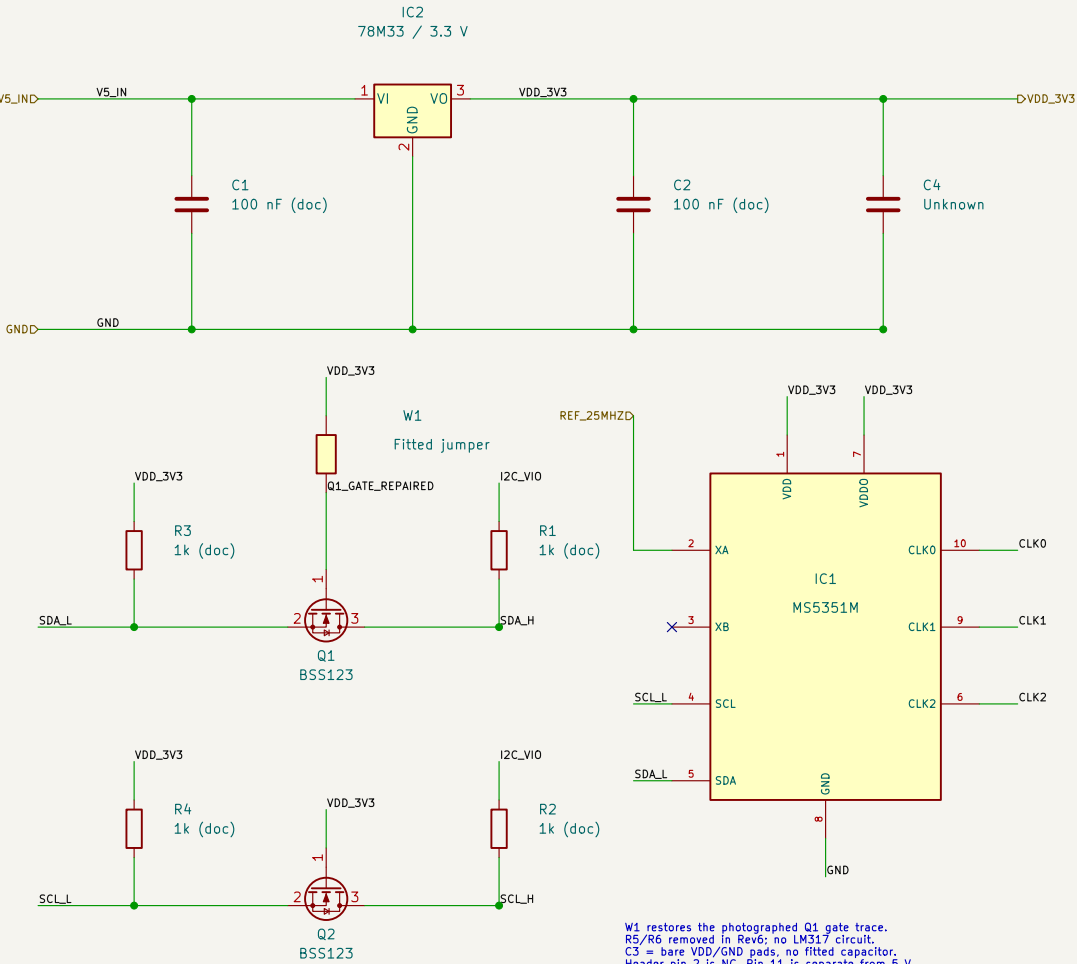
POPULATION AND EVIDENCE

Photos show separate boards; TCXO mounting wires are an intended assembly, not observed solder joints.
Synth pin 11 (I2C_VIO) stays separate from pins 8–10 (5 V input). Supply it for the chosen host bus.
Configure the synthesizer reference as 25,000,000 Hz; omit the 27 MHz crystal.
TCXO capacitor topology is photo-inferred; unknown values and identities remain explicitly unresolved.

Electrical model only; uncertainty and sources in README.md		
QRP Labs circuit; independent documentation model		
Sheet: /		
File: QRP Labs Modules.kicad_sch		
Title: QRP Labs 25 MHz TCXO + Synth Rev6 electrical models		
Size: A4	Date: 2026-10-06	Rev: Model 1
KiCad E.D.A. 10.0.6		Id: 1/3

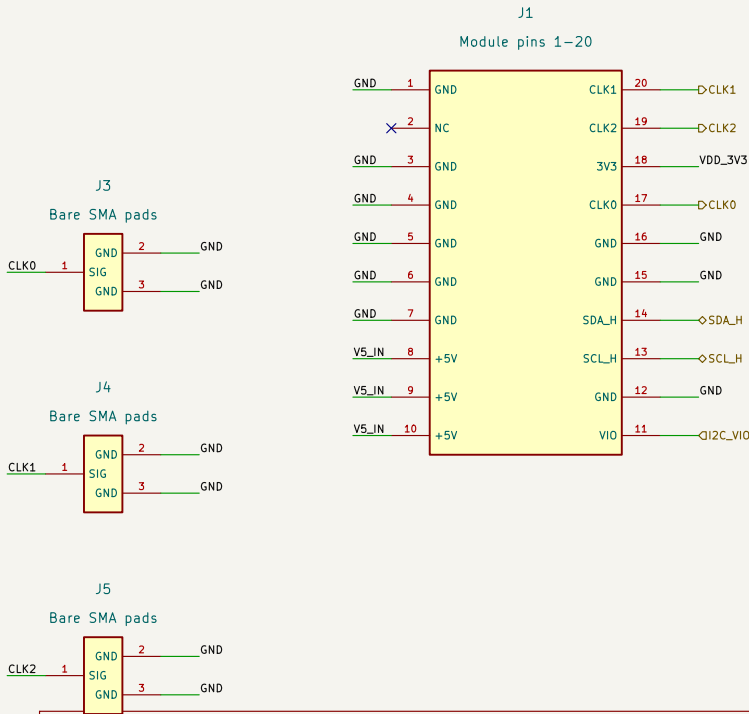
SYNTHESIZER MODULE | 2023 Rev6 | 33 x 25 mm

MS5351M + 78M33 + BSS123. Crystal, direct TCXO, C3, headers and SMA connectors unfitted.



W1 restores the photographed Q1 gate trace.
R5/R6 removed in Rev6; no LM317 circuit.
C3 = bare VDD/GND pads, no fitted capacitor.
Header pin 2 is NC. Pin 11 is separate from 5 V.

Sources: QRP Labs Synth Rev6 manual pp.1–5; Rev6 repair note: Ruimeng MS5351M p3.
C4 value and exact regulator maker unresolved. (doc) = documented value, not a measurement.



Electrical model only; uncertainty and sources in README.md

QRP Labs circuit; independent documentation model

Sheet: /Synthesizer Rev6/

File: Synth Rev6.kicad_sch

Title: QRP Labs synthesizer – photographed Rev6 population

Size: A4

Date: 2026–10–06

Rev: Model 1

KiCad E.D.A. 10.0.6

Id: 2/3

25 MHz TCXO MODULE | 14.8 x 10.5 mm

Three documented module pads; capacitor wiring is photo-inferred.

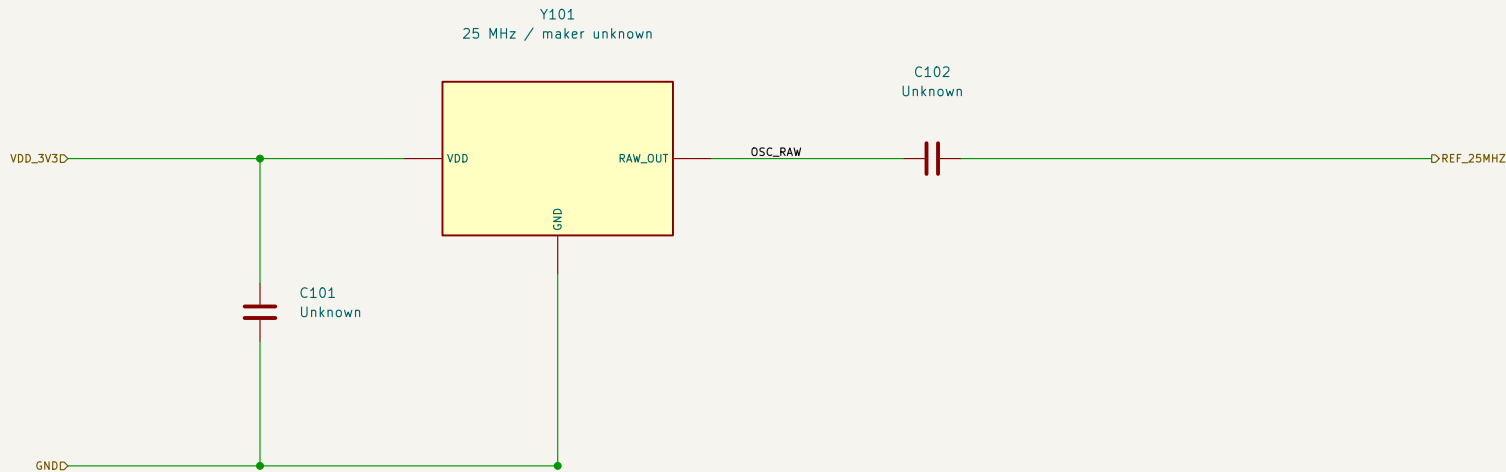


PHOTO-INFERRED INTERNAL DETAIL – verify continuity before reuse

C101: supply bypass. C102: series output coupling. Neither value is published in the checked sources. Y101 uses functional VDD / GND / OUT identifiers, not assumed package pin numbers. The fourth oscillator pad, maker, output waveform and amplitude remain unidentified.

DOCUMENTED INTERFACE

VDD = regulated 3.3 V. OUT = 25 MHz reference. GND = common return.
Sources: QRP Labs TCXO page and Synth Rev6 manual pp.5–6; photos IMG_1702 / IMG_1705.

Electrical model only; uncertainty and sources in README.md		
QRP Labs circuit; independent documentation model		
Sheet: /25 MHz TCXO/ File: TCXO 25MHz.kicad_sch		
Title: QRP Labs 25 MHz TCXO daughterboard		
Size: A4	Date: 2026–10–06	Rev: Model 1
KiCad E.D.A. 10.0.6		Id: 3/3